

YANAN HE

📍 New Haven, CT 🌐 <https://echoo113.github.io>

I study why AI works when it works, why it fails when it doesn't, and how understanding those differences can help us build better models.

EDUCATION

Yale University, New Haven, CT Aug 2026 – May 2028
M.S. in Computer Science (*Fully Funded*)

Purdue University, West Lafayette, IN Aug 2023 – May 2026
B.S. in Computer Science & Mathematics-Statistics GPA: 3.96/4.0 | *Graduated with Distinction*

RESEARCH EXPERIENCE

NLP Lab, Yale University *New Haven, CT*
Research Assistant | Advisor: Arman Cohan Apr 2026 – Present

- LLM Post-Training – studying existing methods (e.g., OPD, SDPO) and developing new RL methods.
- LLM Evaluation, Alignment, and Long-Horizon Agents [1]

Graph Lab, Stony Brook University *Stony Brook, NY*
Research Assistant | Advisor: Tengfei Ma May 2025 – Apr 2026

- Time-Series Applications – JEPA-style architectures [2] and causal multi-scale modeling [3]
- LLM Reasoning in Geometric Space – hyperbolic latent navigation [4]

SELECTED PUBLICATIONS

1. Time to REFLECT: Can We Trust LLM Judges for Evidence-based Research Agents? * Equal contribution.
Leyao Wang*, **Yanan He***, Peng Chen, Asaf Yehudai, Yixin Liu, Rex Ying, Michal Shmueli-Scheuer, Arman Cohan
arXiv:2605.19196.
2. SC-JEPA: Stabilizing Latent Predictive Learning for Time-Series Anomaly Prediction
Yanan He, Yunshi Wen, Xin Wang, Tengfei Ma
SDM 2026; Forecast@ICML26.
3. CAAD: Causality-Aware Multivariate Time Series Anomaly Detection via Multi-Scale Alignment and Structural Causal Consistency
Xin Wang, Yunshi Wen, **Yanan He**, Haotian Xu, Youlan Zhao, Michel Ferreira Cardia Haddad, Tengfei Ma
SIGKDD 2026 Research Track.
4. HyperGuide: Hyperbolic Guidance for Efficient Multi-Step Reasoning in Large Language Models
Yuyu Liu, Haotian Xu, **Yanan He**, Sarang Rajendra Patil, Mengjia Xu, Tengfei Ma
arXiv:2605.24140.

INTERESTS & SKILLS

- **Programming Languages:** Python, Java, C/C++, Assembly, R, SQL, MATLAB, JavaScript, HTML/CSS
- **Languages:** English (Professional), Chinese (Native)